

MICHAEL C. PUENTE

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MECHANICAL DESIGN | MANUFACTURING | THERMAL-FLUIDS | HVAC/MEP

ENGINEERING PROFILE

Mechanical Engineering student with hands-on experience in mechanical design, machine-element analysis, commercial HVAC and plumbing documentation, aerospace CFD, equipment handling, and vehicle diagnostics. Applies SolidWorks, MATLAB, OpenFOAM, AutoCAD, Bluebeam Revu, and Excel across design, analysis, and field-supported work. Seeking internship or co-op opportunities in mechanical design, manufacturing, aerospace, thermal-fluid systems, or reliability engineering.

EDUCATION

The University of Texas at San Antonio (UTSA) - B.S. Mechanical Engineering

Expected Fall 2027

Relevant Coursework: Manufacturing Engineering, Machine Element Design, Measurements and Instrumentation, Mechanics of Solids, Numerical Methods, Dynamic Systems and Controls

Texas Lutheran University - Applied Physics Coursework

Activities: TLU Varsity Football | American Society of Mechanical Engineers | Society of Physics Students

TECHNICAL SKILLS

CAD and Mechanical Design: SolidWorks parts, assemblies, drawings, and Flow Simulation; AutoCAD; Revit; AGMA gear analysis; shaft, bearing, and key design; V-M-T loading; Modified Goodman fatigue analysis

Analysis and Computing: MATLAB/Simulink, MATLAB beam-element FEM, OpenFOAM, ParaView/ParaFoam, Python/Jupyter, Linux

Engineering and Field Tools: Bluebeam Revu, Excel, drawing and specification review, quantity takeoffs, equipment verification, measurement and instrumentation, calipers, multimeters, hand and power tools

ENGINEERING AND TECHNICAL EXPERIENCE

Mechanical Engineering Intern | Sur Texas LLC | San Antonio, TX

Jan. 2025 - Present

- Review commercial HVAC and plumbing bid drawings and perform quantity takeoffs for ductwork, piping, equipment, fixtures, and accessories using Bluebeam Revu and Excel.
- Produced Delta 1 plumbing revisions across permit sheets P.3-P.9 after architectural floor-plan changes; added and relocated fixtures and equipment, revised sanitary, vent, and domestic-water routing, coordinated risers, details, and schedules, and removed obsolete work.
- Track revisions, scope changes, and bid updates to improve material coverage, technical clarity, and estimating accuracy across commercial projects.

Undergraduate Researcher | Klesse Undergraduate Research Program, UTSA

Summer 2026

- Conducted OpenFOAM simulations of Mach 6 turbulent boundary-layer flow over adiabatic flat-plate and three-dimensional cavity geometries; evaluated pressure, skin friction, thermal behavior, and flow fields.
- Built meshes and boundary conditions, executed parallel jobs on TACC and ARC HPC systems, and interpreted results in ParaView/ParaFoam.

Warehouse Associate | Texas Air Products | San Antonio, TX

Jul. 2026 - Present

- Receive, verify, identify by tag and application, stage, and load commercial HVAC equipment and components for contractor orders.
- Build hands-on equipment knowledge across dampers, air-distribution devices, controls, VFDs/starters, curbs, isolators, ductwork, and accessories.

SELECTED ENGINEERING PROJECTS

Two-Stage Spur-Gear Speed Reducer | Team Machine Element Design Project

SolidWorks | MATLAB

- Designed an 82-hp, two-stage spur-gear transmission reducing 1,200 rpm to 100 rpm through 4:1 and 3:1 stages for an overall 12:1 reduction.
- Performed AGMA bending and contact analysis, V-M-T loading, Modified Goodman fatigue analysis, MATLAB beam-element FEM, and bearing and key selection; created the complete SolidWorks assembly and dimensioned shaft drawings.

Arduino DC Motor Speed Control | Team Laboratory Project

Arduino | Instrumentation

- Built and tested a PWM motor-control circuit with potentiometer input, Arduino Uno, 2N2222 transistor, flyback diode, and LCD; four trials per operating point produced a 743.6-8,225 rpm mean speed range and exposed a low-voltage switching limit.

Heat Exchanger and Jet Engine Rear Casing CAD Studies

SolidWorks

- Recreated a tutorial-guided shell-and-tube heat exchanger and independently corrected the assembly so shell-side and tube-side flow paths remained separate.
- Modeled a work-in-progress jet-engine rear casing as a single-component design exercise using revolve, boss-extrude, sweep, cut-extrude, fillets, circular patterns, and face edits.

1994 Ford F-150 Mechanical Restoration

Hands-On Project

- Returned a non-running truck to operation through fuel, ignition, starting, electrical, and mechanical diagnostics, troubleshooting, and component replacement.